Product Requirements Document (PRD)

> **Project Name:** TrailPulse
>
> **Type:** Offline Hyperlocal Nature Audio Guide (PWA + Express Orchestrator)
>
> **Hackathon:** Hacktoberfest 2026 — Week 2 Challenge ("Touch Grass")
>
> **Target Categories:** Best Use of Gemma ($200), Best Use of ElevenLabs ($100), Best Use of Render ($200) / Best Use of DigitalOcean ($200)

1. Executive Summary & Problem Statement

1.1 Background

Modern outdoor technology suffers from a paradox: apps intended to enhance time in nature (plant identifiers, digital trail maps, peak finders) pull the user's eyes directly toward glass screens. Hikers stop at vistas, open apps, scan menus, and read paragraphs of text while ignoring the physical landscape around them. Furthermore, backcountry recreation frequently occurs in complete cellular dead zones where conventional cloud-hosted AI APIs fail entirely.

1.2 Core Thesis

TrailPulse inverts this interaction model by treating the screen as an obstacle to be minimized. By leveraging open-weight models (Gemma) and high-fidelity speech synthesis (ElevenLabs) before heading outdoors, TrailPulse packages dynamic, context-aware audio tours that execute **100% offline via local device GPS**. The hiker keeps their phone in their pocket, listening to hands-free narration triggered dynamically as they walk past geographical points of interest.

2. Target Persona & User Stories

2.1 Target Persona

- **Name:** Trail Runner / Day Hiker Sarah
- **Profile:** Enjoys day hikes and forest trails, wears wireless earbuds, values solitude, dislikes stopping to inspect phone screens, and frequents trails with poor or non-existent cellular connectivity.

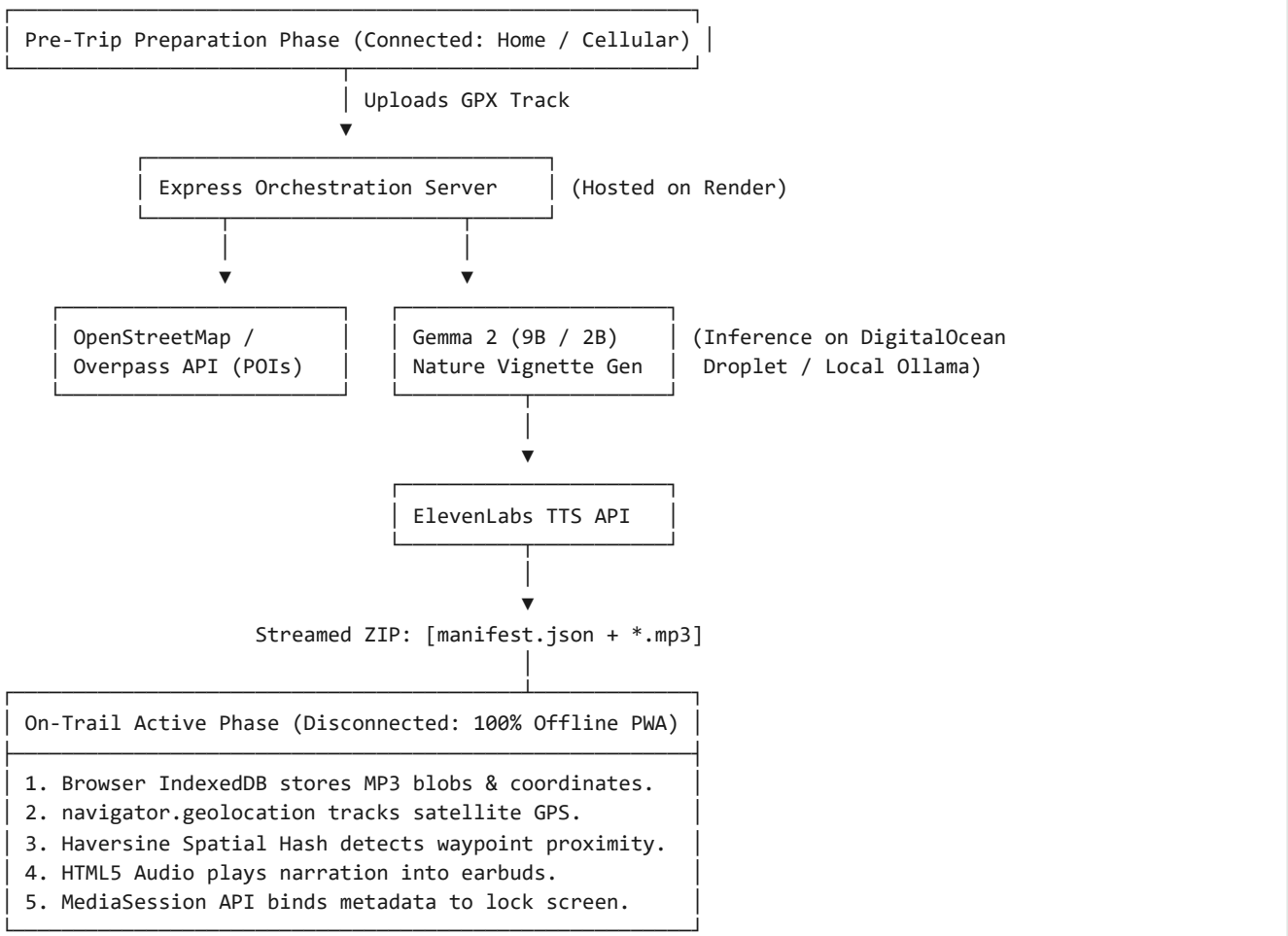
2.2 User Stories

- **US-1 (Pre-Trip Planning):** As a hiker on home Wi-Fi, I want to upload a GPX route or select a trail so that I can automatically generate and cache an audio tour bundle before leaving cell service.
- **US-2 (Hands-Free Narration):** As a hiker on the trail, I want contextual audio vignettes to trigger automatically through my earbuds when I approach landmarks, so that I never have to pull my phone from my pocket.
- **US-3 (Zero-Signal Resilience):** As a backcountry trekker in airplane mode, I want all geofences, audio tracks, and location tracking to run without network access so that poor signal does not disrupt the experience.

- **US-4 (No Nuisance Loops):** As a hiker backtracking along a trail or navigating switchbacks, I do not want the same audio track to trigger repeatedly in a short window.
- **US-5 (Lock Screen Control):** As a user with a locked phone, I want to see the landmark name on my lock screen and pause/replay narration using earbud tap controls.

3. Product Architecture & Technical Scope

3.1 System Topology



3.2 Technology Stack

- **Client App:** Vite + React PWA (Single Page Application, installable to iOS/Android).
- **Client Storage:** IndexedDB via `idb-keyval` (storing audio Blobs) and `fflate` (in-memory ZIP extraction).
- **Client APIs:** HTML5 Geolocation API, HTMLMediaElement, Media Session API.
- **Backend:** Node.js + Express (stream-based pipeline with `multer` and `archiver`).
- **Open-Source AI Engine:** Gemma 2 (9B-Instruct) via Ollama/vLLM.
- **Audio Synthesis:** ElevenLabs Multilingual V2 (`turbo-v2`).
- **Geospatial Extraction:** OpenStreetMap Overpass API + `@tmcw/togeojson` .
- **Hosting:** Render Web Service (monorepo serving static client bundle + API endpoints).

4. Functional Specifications

4.1 GPX Ingestion & Waypoint Selection

- Accepts standard `.gpx` files up to 10 MB.
- Extracts coordinates, calculates total trail distance, and samples representative coordinates (trailhead, elevation peaks, midpoints, and terminus).
- Queries Overpass API within a 500m radius of sampled points for natural (`natural=*`), historic (`historic=*`), or scenic features (`tourism=viewpoint`).
- Caps total waypoints to 5–8 high-value trigger stops per 5 miles of trail to prevent audio fatigue.

4.2 Gemma Prompt Engineering & Vignette Synthesis

- **Persona:** Field naturalist.
- **Length Constraint:** 45–60 spoken words (20–30 seconds of audio).
- **Sensory Grounding:** Strict instruction to reference tactile, visual, or acoustic features (e.g., moss, tree bark, stone strata, bird canopy) rather than dry encyclopedic dates.
- **Negative Guardrails:** Never begin with conversational filler ("Hello hiker", "Welcome to...").

4.3 Offline Asset Packaging

- Audio clips are compressed to 64 kbps mono MP3 files.
- A structured `trail_manifest.json` is created containing waypoint IDs, coordinates, trigger radii, cooldown limits, and audio references.
- The server compiles the bundle into a `.zip` archive and streams it directly to the client without persistent intermediate disk storage.

4.4 Geofence & Playback Engine (Client-Side)

- **Trigger Radius:** 25 meters around target coordinates.
- **Geodesic Calculation:** Client computes distance via the Haversine formula on every location update.
- **Cooldown & Hysteresis:** Each waypoint enforces a 15-minute (900-second) cooldown to prevent re-triggering while resting or looping back on switchbacks.
- **Audio Concurrency:** Only one audio clip plays at a time. New trigger events are ignored if an existing track is active.

5. Non-Functional Requirements

Metric	Target	Rationale
Offline Capability	100% operation in Airplane Mode	Trailheads lack cellular reception; zero network requests allowed after bundle download.
Client Bundle Size	< 25 MB per 10-mile trail	Accommodates mobile IndexedDB quotas and fast local extraction.
GPS Battery Efficiency	< 8% battery drain per hour	Dynamic GPS polling intervals (30s baseline, stepping down to 3s when within 100m of a POI).
Audio Latency	< 150ms trigger-to-play	Instant playback upon entering the 25m boundary.
Screen Time	< 60 seconds total screen-on time	User initiates trail mode, puts phone away, and navigates purely by ear.

6. Success Metrics & Hackathon Judging Alignment

6.1 Hackathon Scoring Alignment

- **Writing Quality (Top Weight):** The README and Dev.to post will articulate the philosophical argument: technology should connect humans with the physical world, not mediate it through screens.
- **Relevance to Theme ("Touch Grass"):** Audio keeps eyes off screens and on trails. The submission includes real field-test photos and GPS traces.
- **Why Open Innovation Matters:**
  - **Edge Privacy:** Location traces never leave the hiker's phone while outdoors.
  - **Zero Cloud Dependency:** Open weights (Gemma) allow local model compilation and reproducible script generation without closed SaaS paywalls.
  - **Community Geodata:** Combines open weights with OpenStreetMap public commons data.

6.2 Milestones & Deliverables

1. **Milestone 1:** Node.js backend parsing GPX, querying Overpass, prompting Gemma, and calling ElevenLabs.
2. **Milestone 2:** Streaming `.zip` bundler and client-side unzipper via `fflate` and `idb-keyval`.
3. **Milestone 3:** Geofence audio engine with Media Session API background play.
4. **Milestone 4:** Monorepo deployment to Render; physical trail test with photos and DevRelay/Sentry tracing.